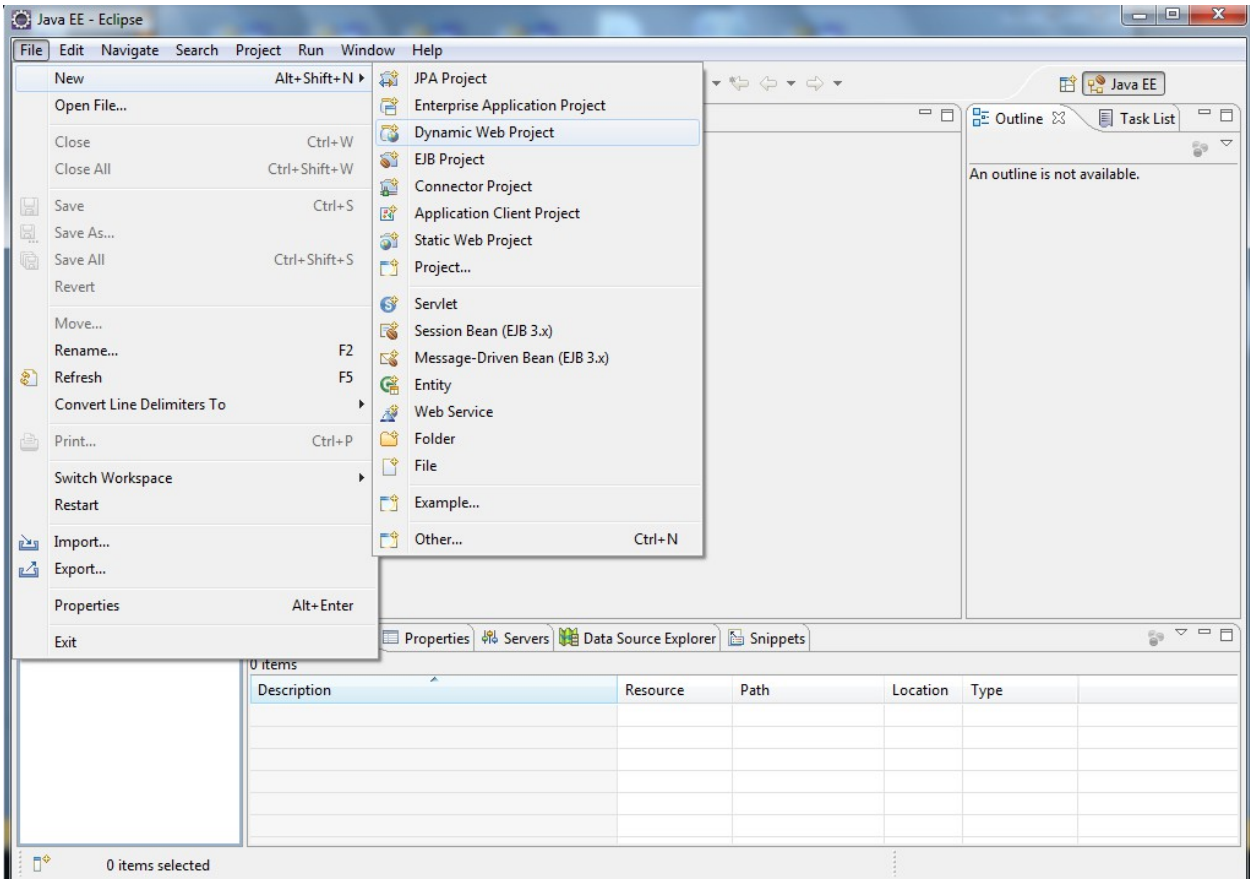
Introduction to Servlet, Java Web Project and Web.xml

Problem Statement

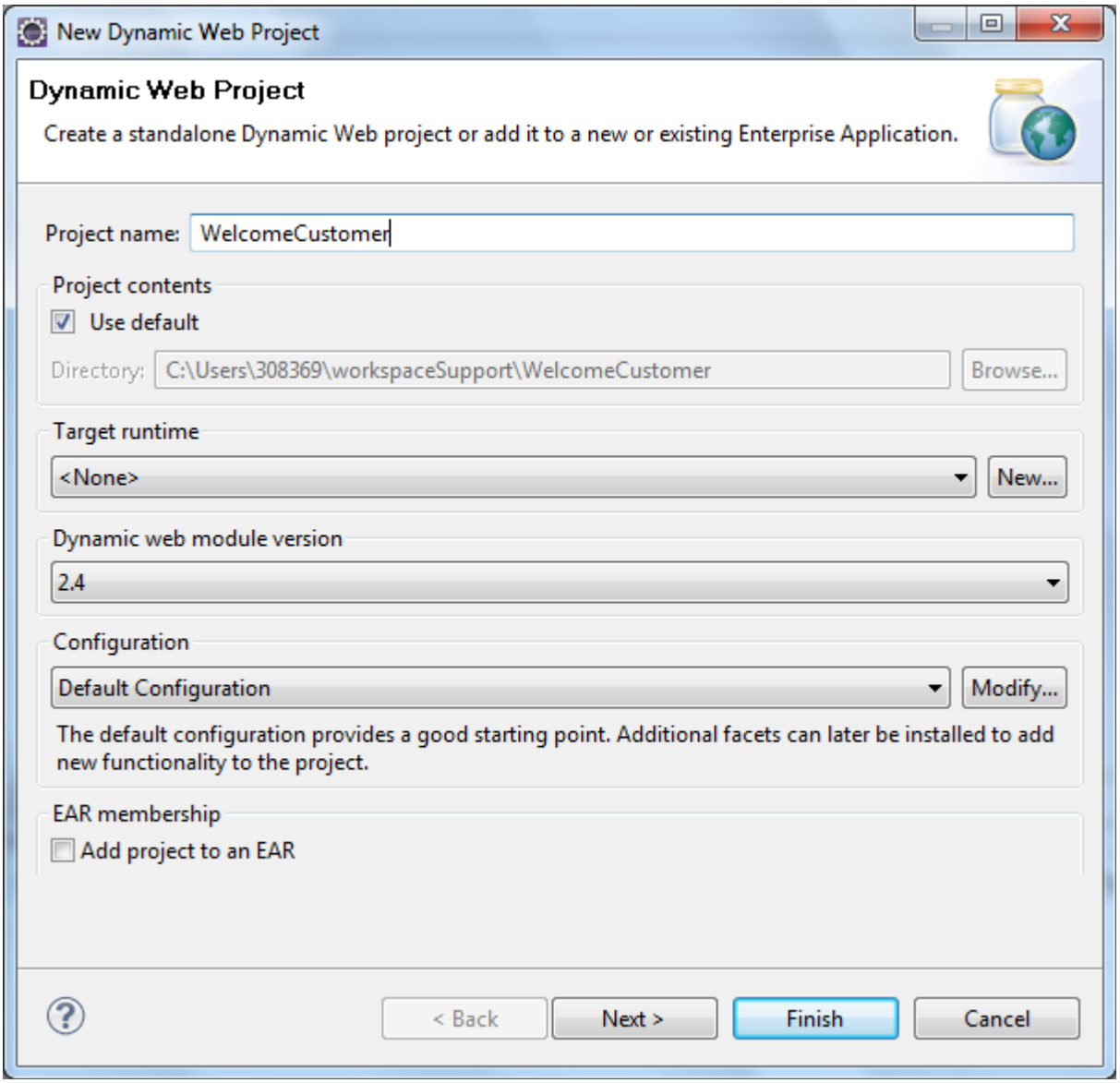
Develop a simple java web application to get a Customer name and show a welcome message to the user when he submits the form.
Use HTML for developing view component.

1) WelcomeCustomer

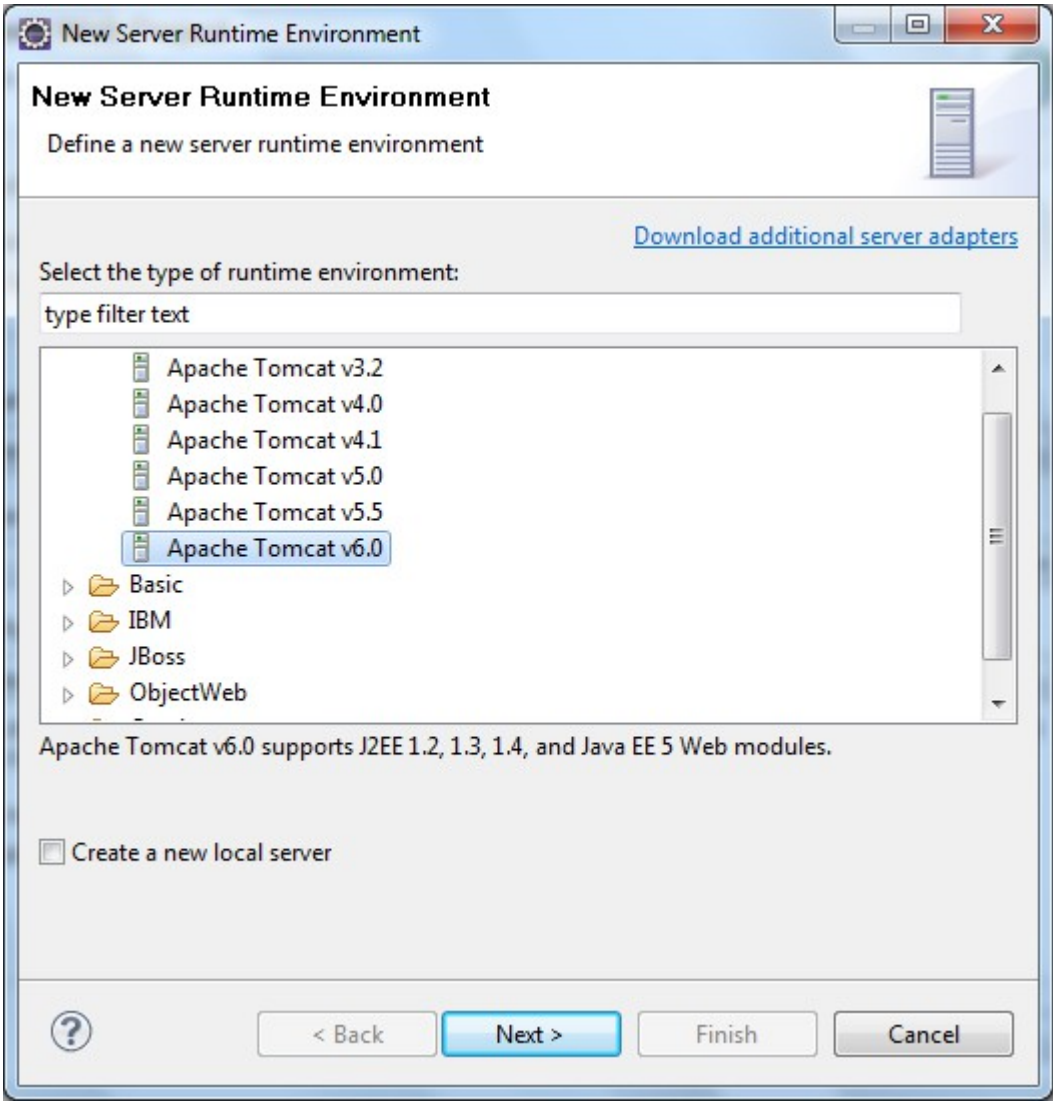
Create dynamic Web project **by File->New->Dynamic Web Project**



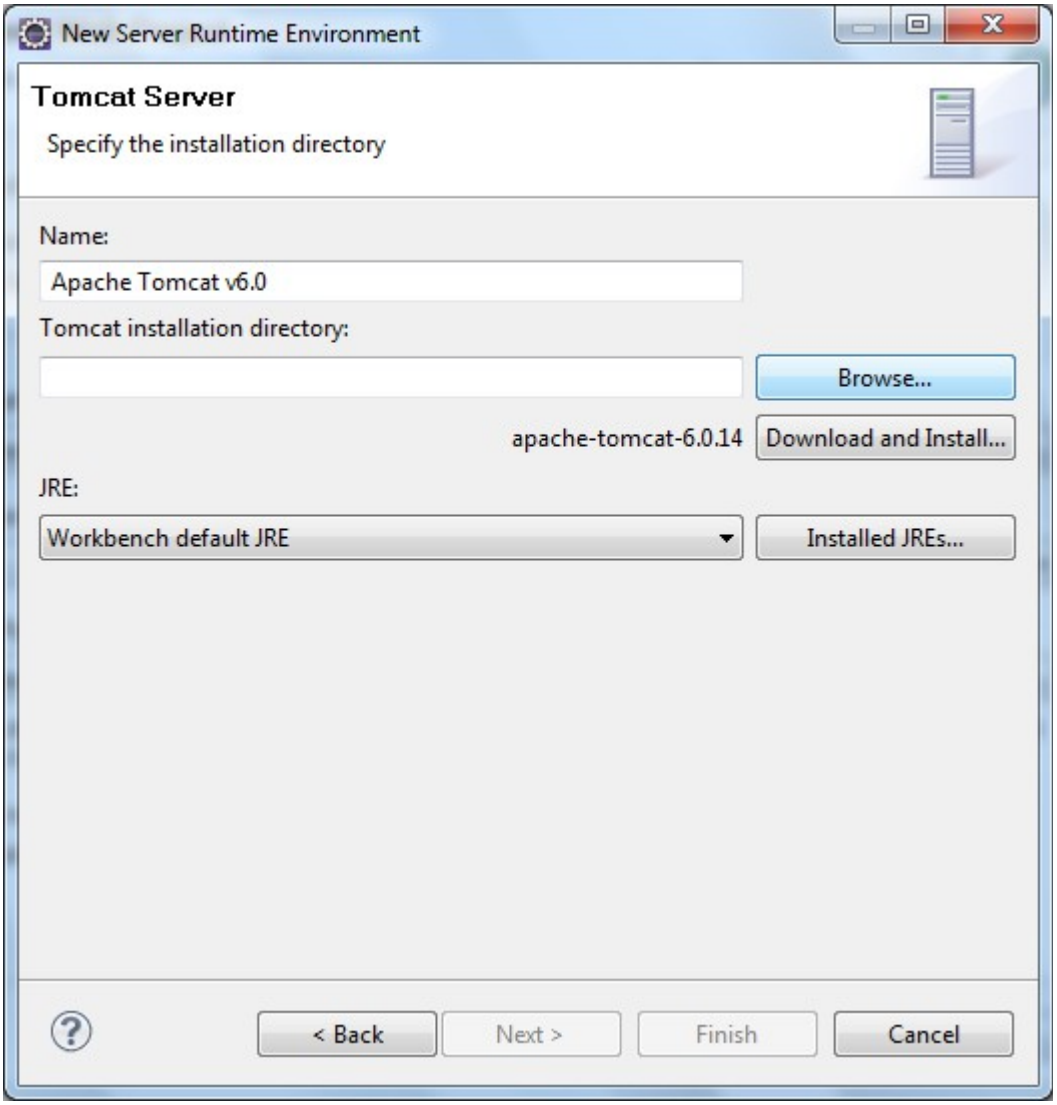
Provide a valid project name "WelcomeCustomer" adhering to the coding standards



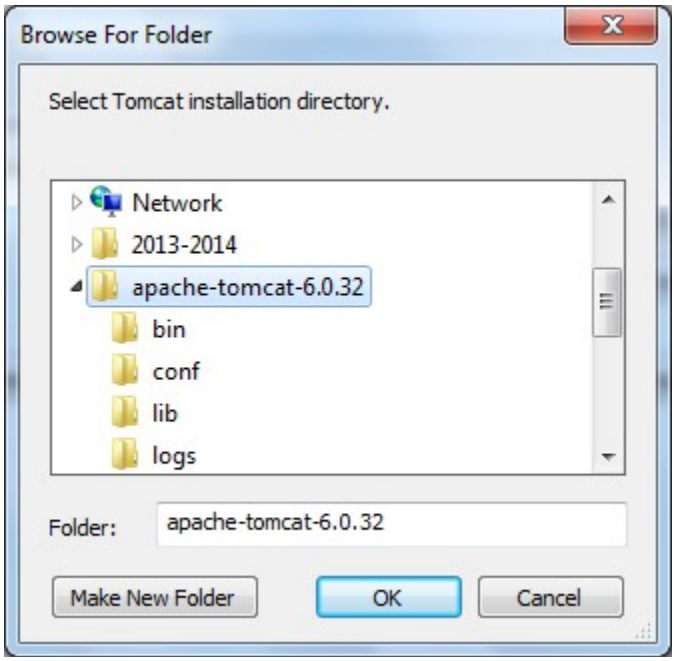
Set the new target runtime environment by clicking on New in Target runtime. Select **Apache Tomcat v6.0** and click on Next button.



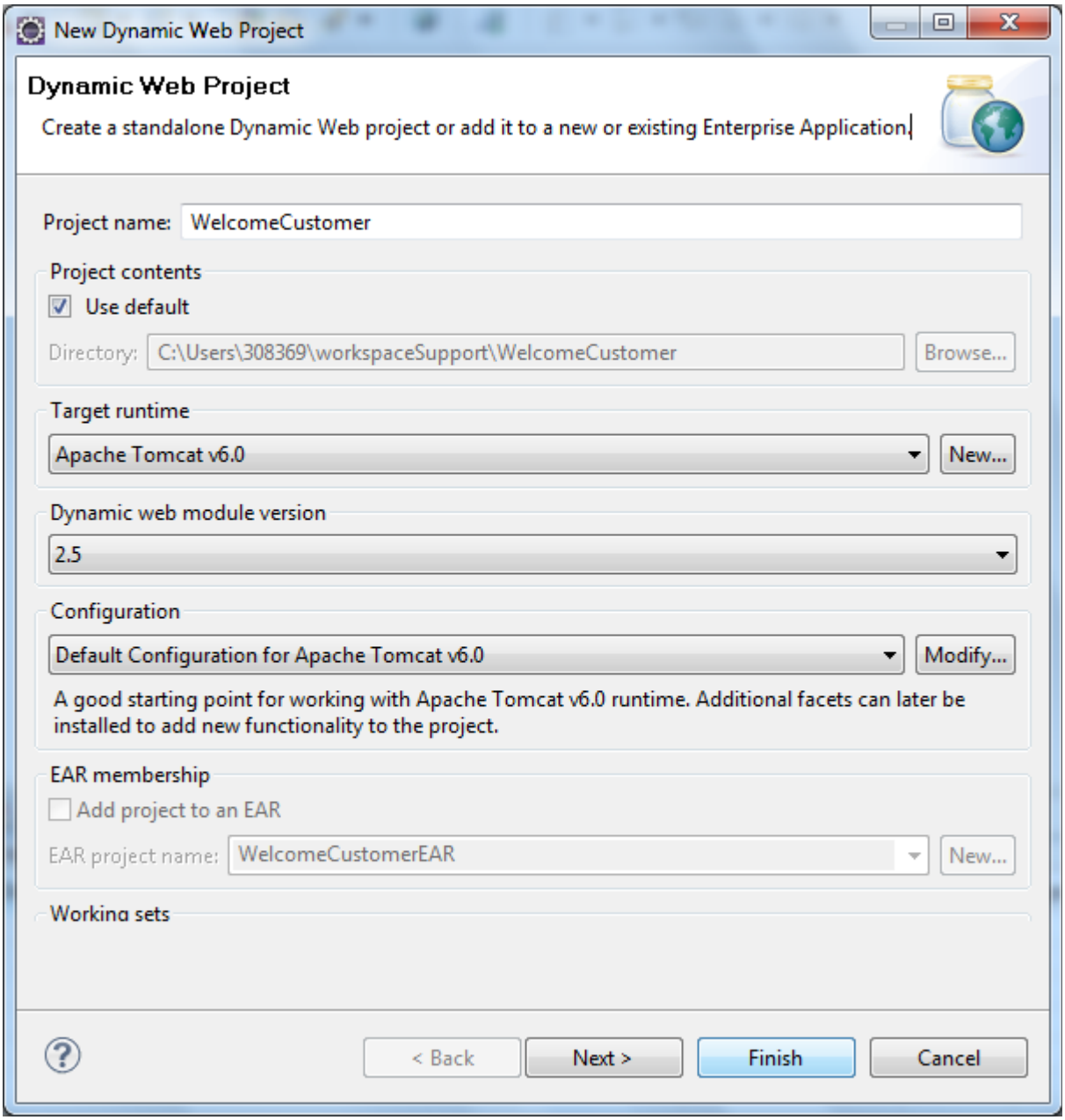
Browse for the right installation directory for Apache Tomcat v6.0



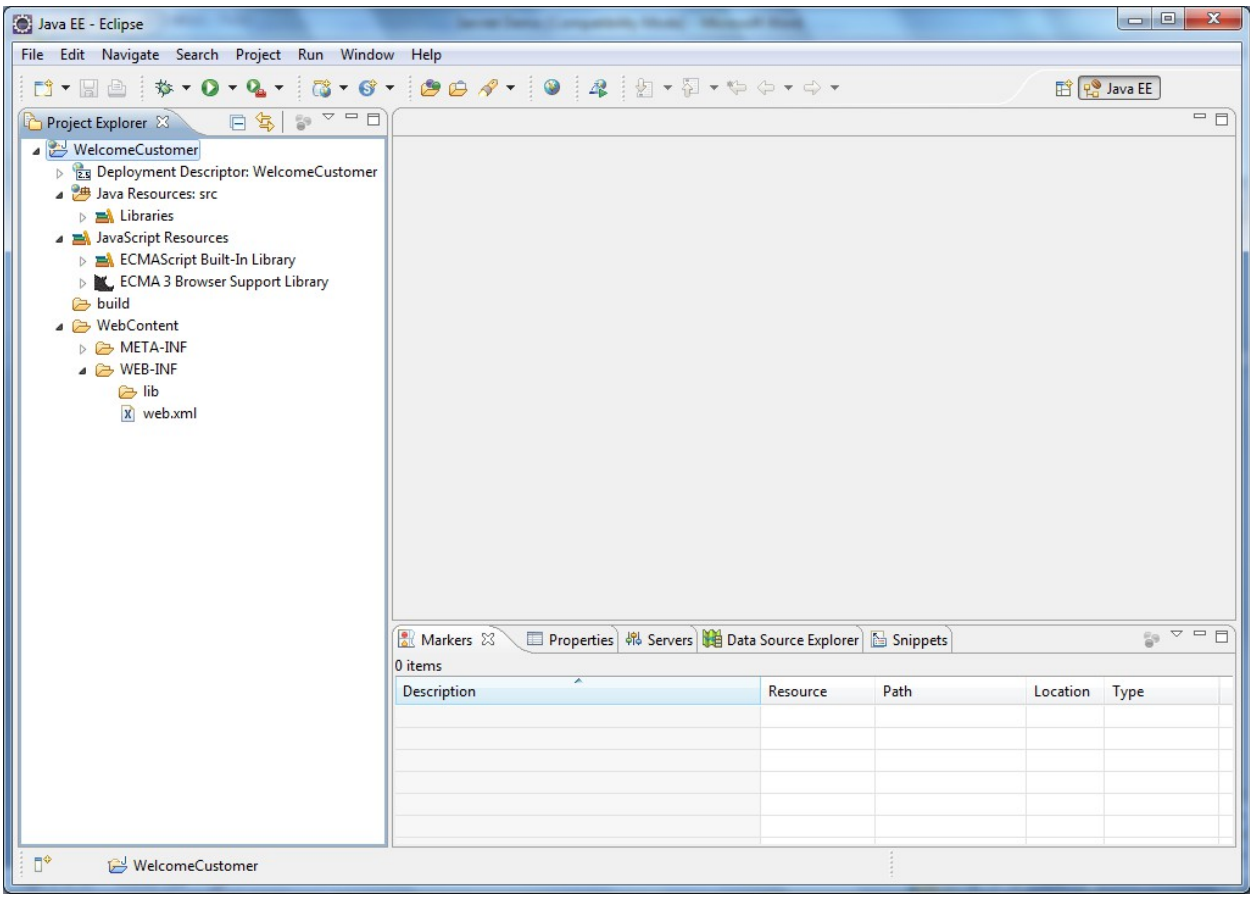
Browse for the right folder of apache-tomcat-6.032 from the local system and click on OK



Once the target runtime gets populated click on finish.

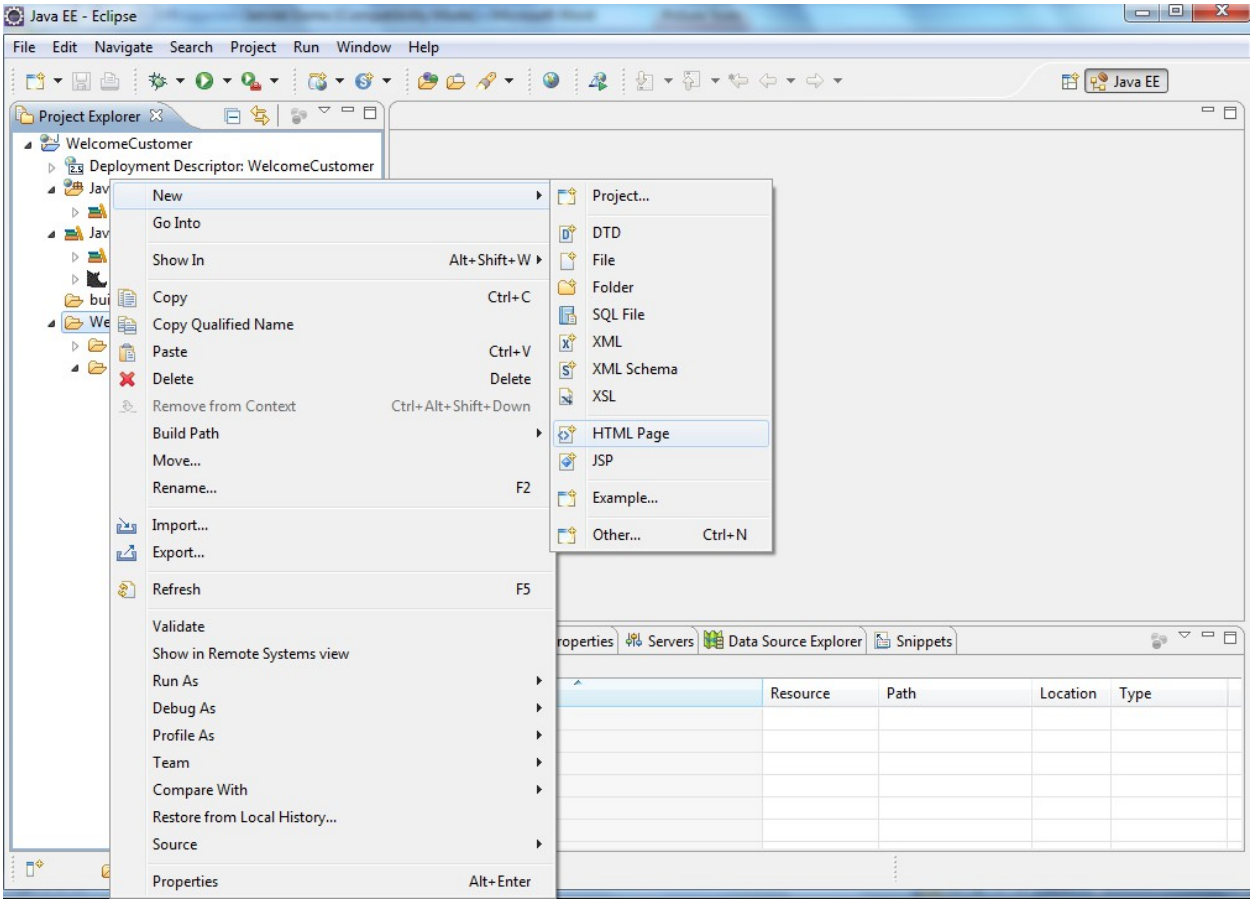


A dynamic Web Project “WelcomeCustomer” with the folder structure as follows gets created. The same can be viewed in the project explorer.

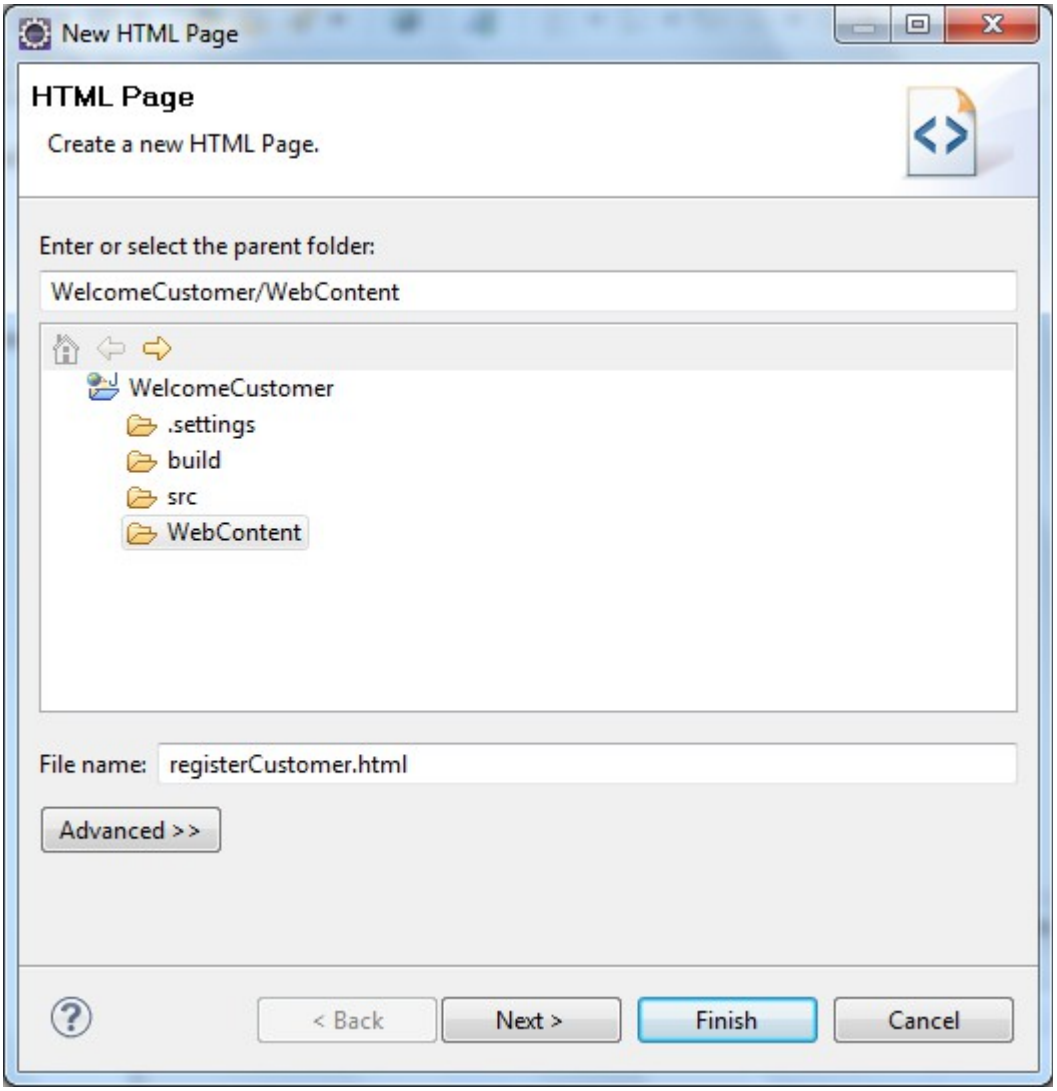


2) registerCustomer.html

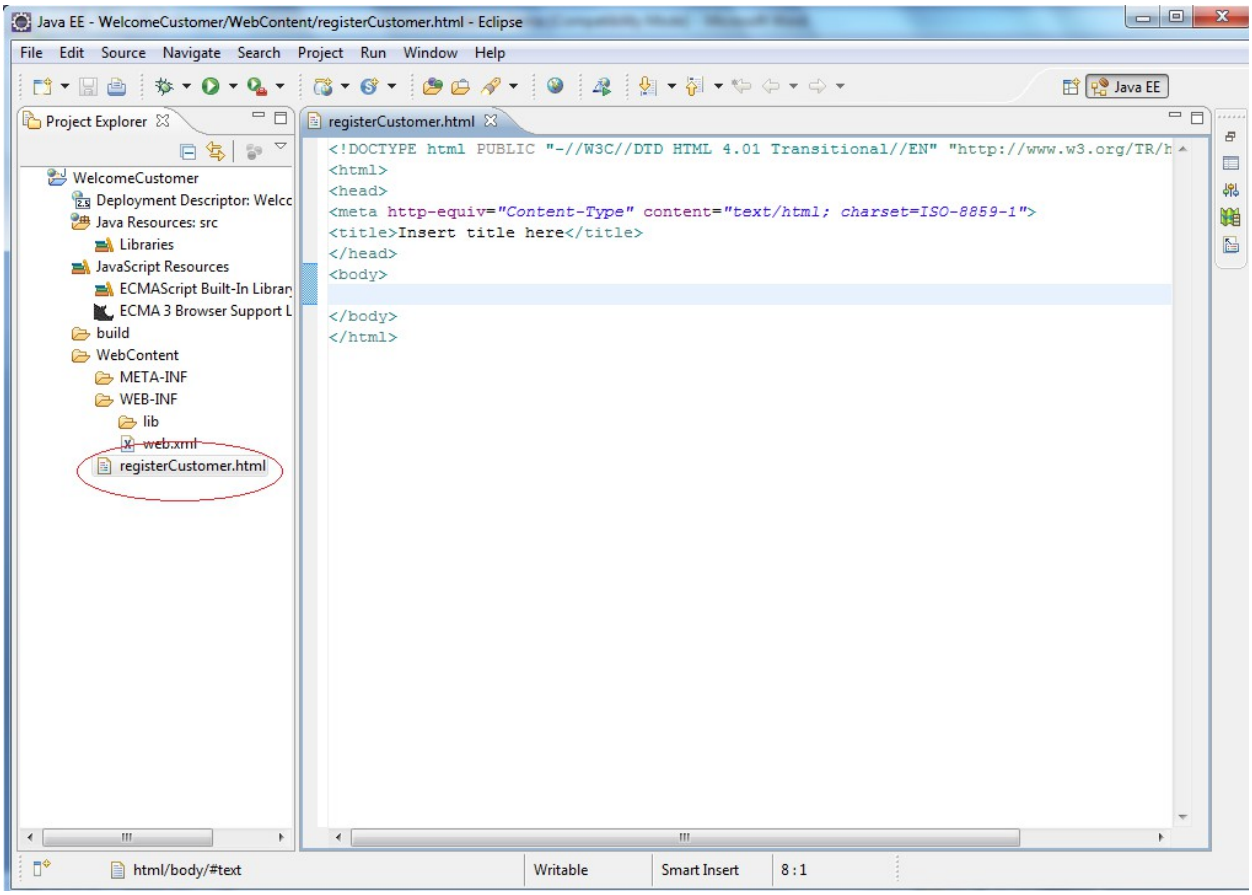
Create registerCustomer.html inside Webcontent of the Dynamic Web project WelcomeCustomer.



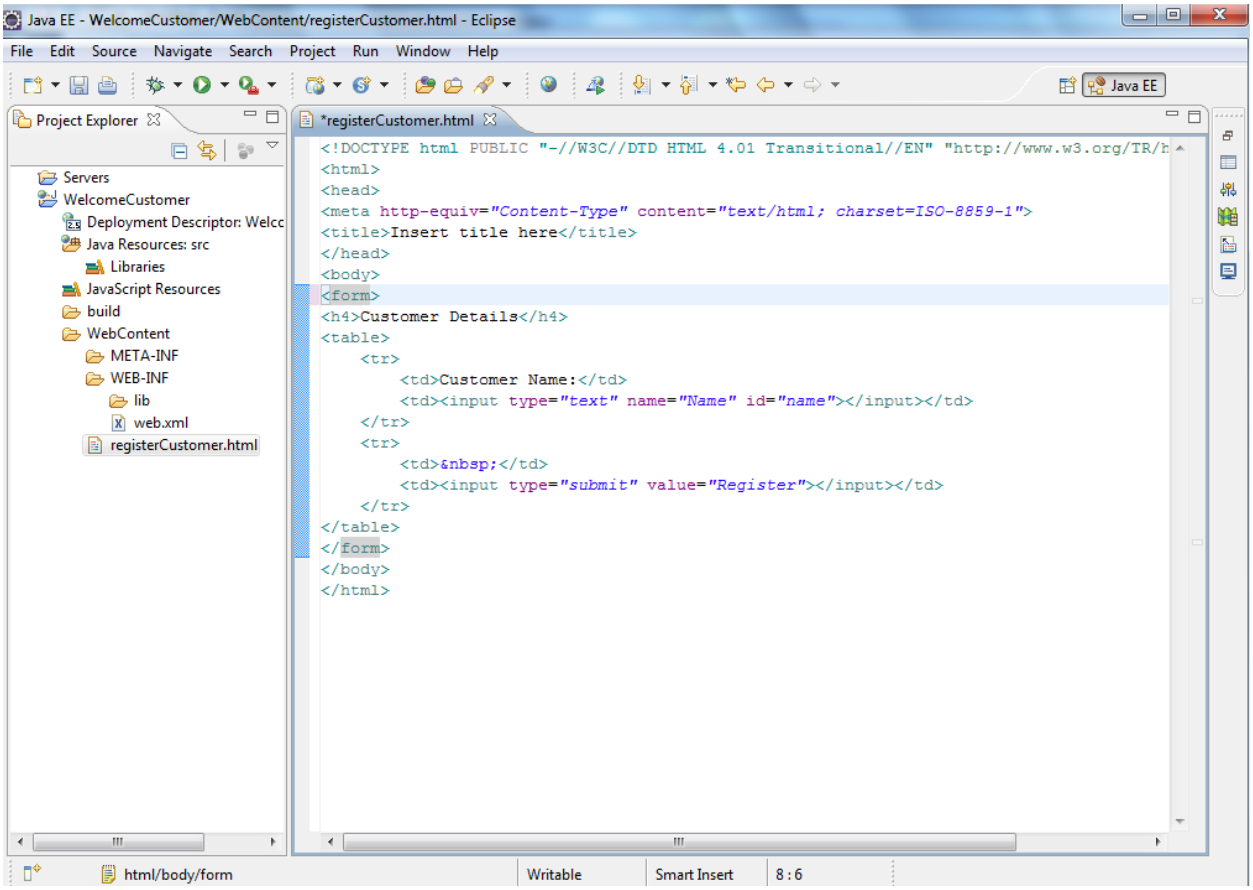
Provide a valid name for the html page and click on ‘Finish’.



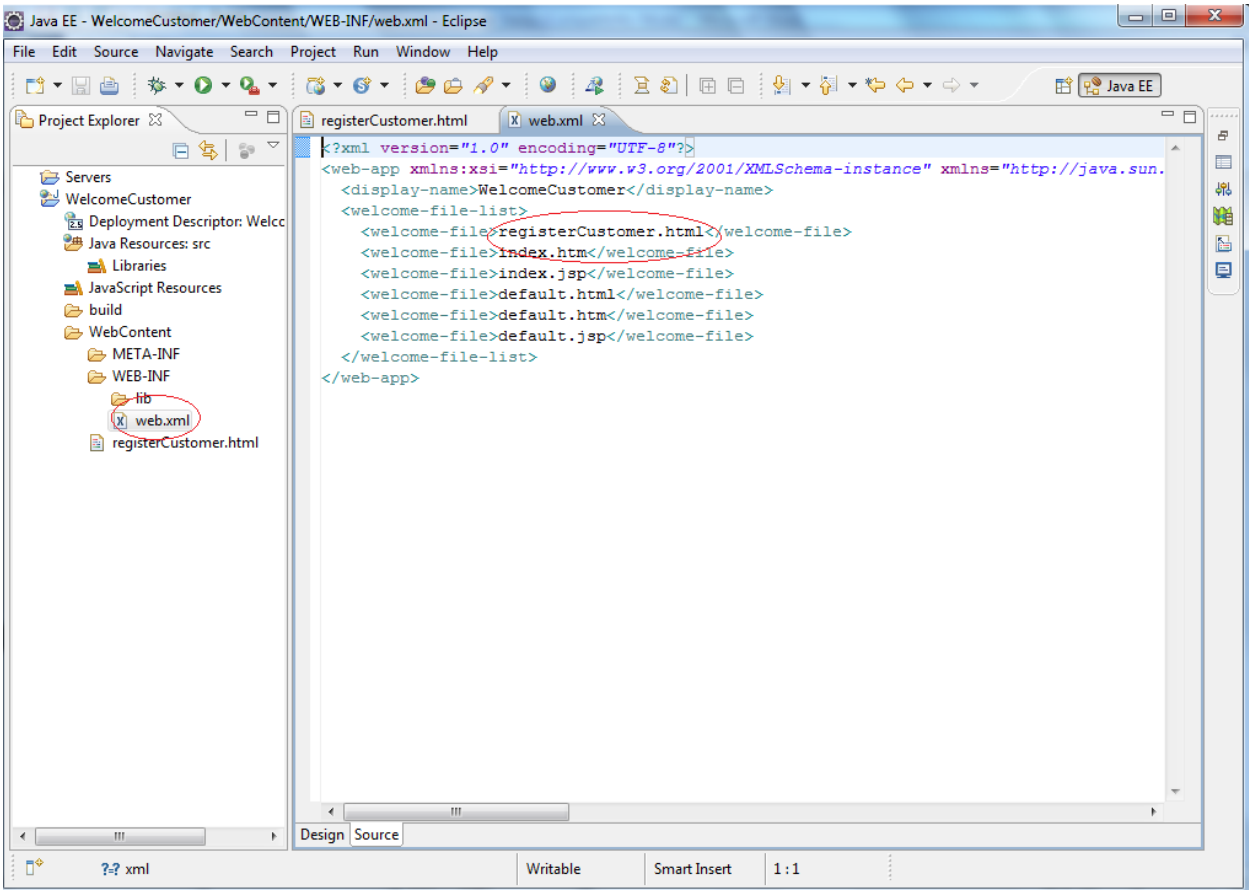
“registerCustomer.html” gets created inside the WebContent folder as highlighted



Create a form inside the registerCustomer.html that will get the Customer name.

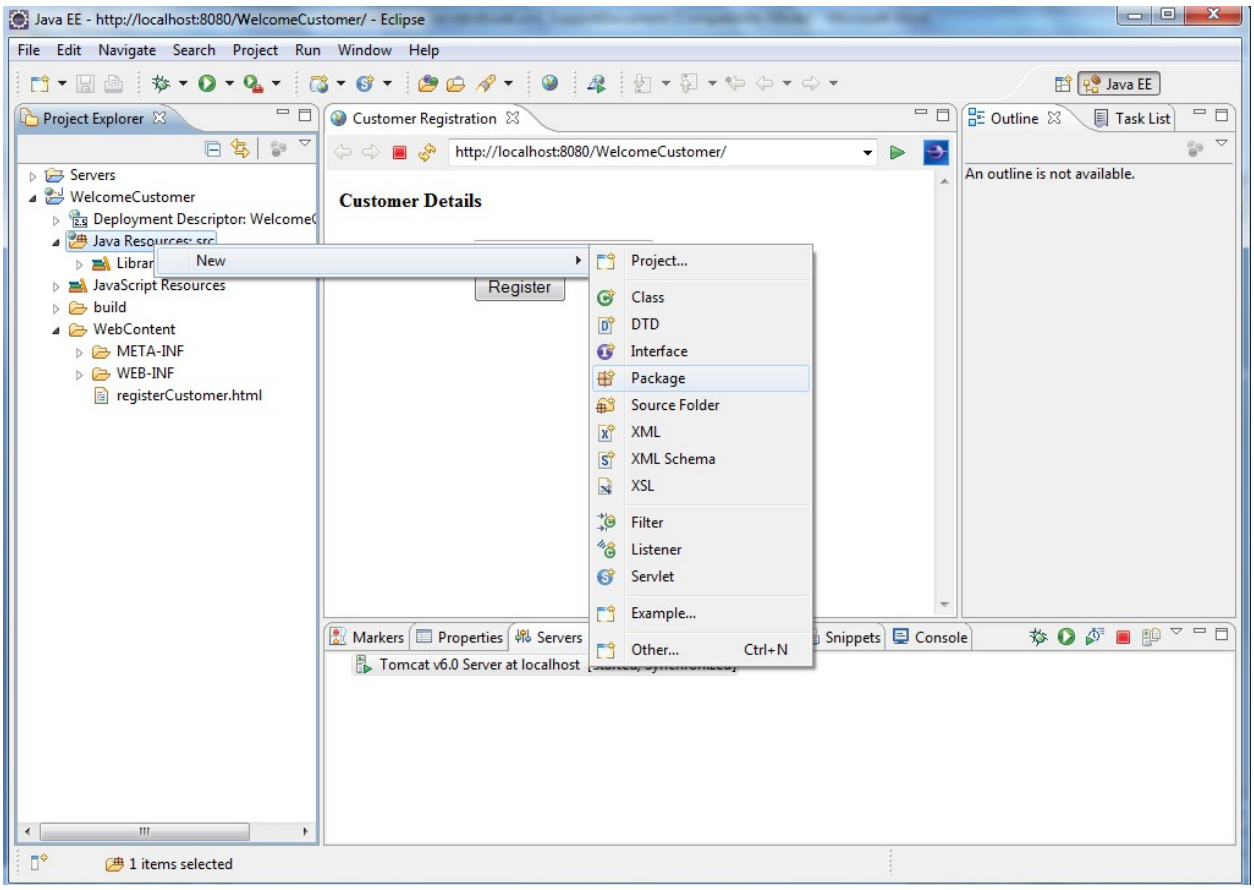


Provide the welcome file name as registerCustomer.html inside **web.xml**

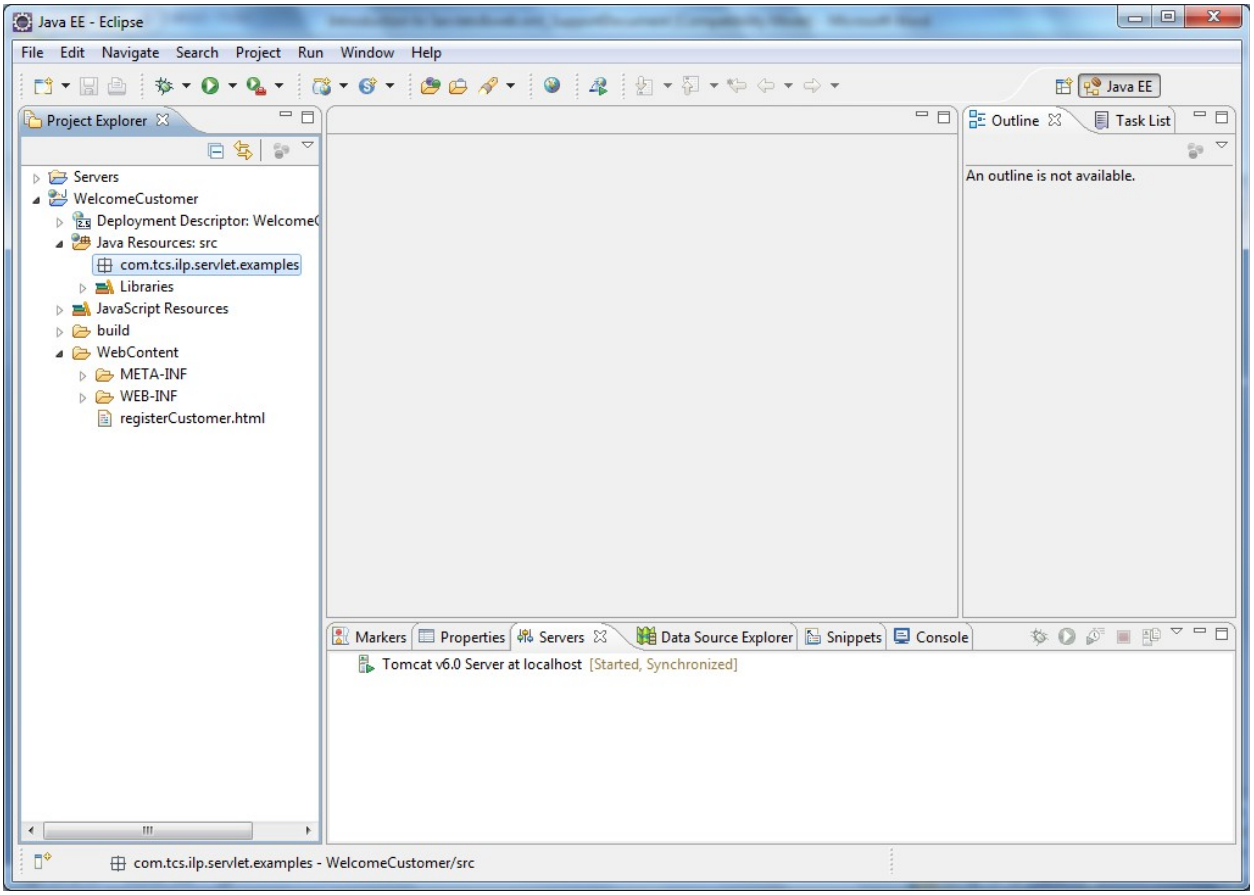


3) CustomerController.java

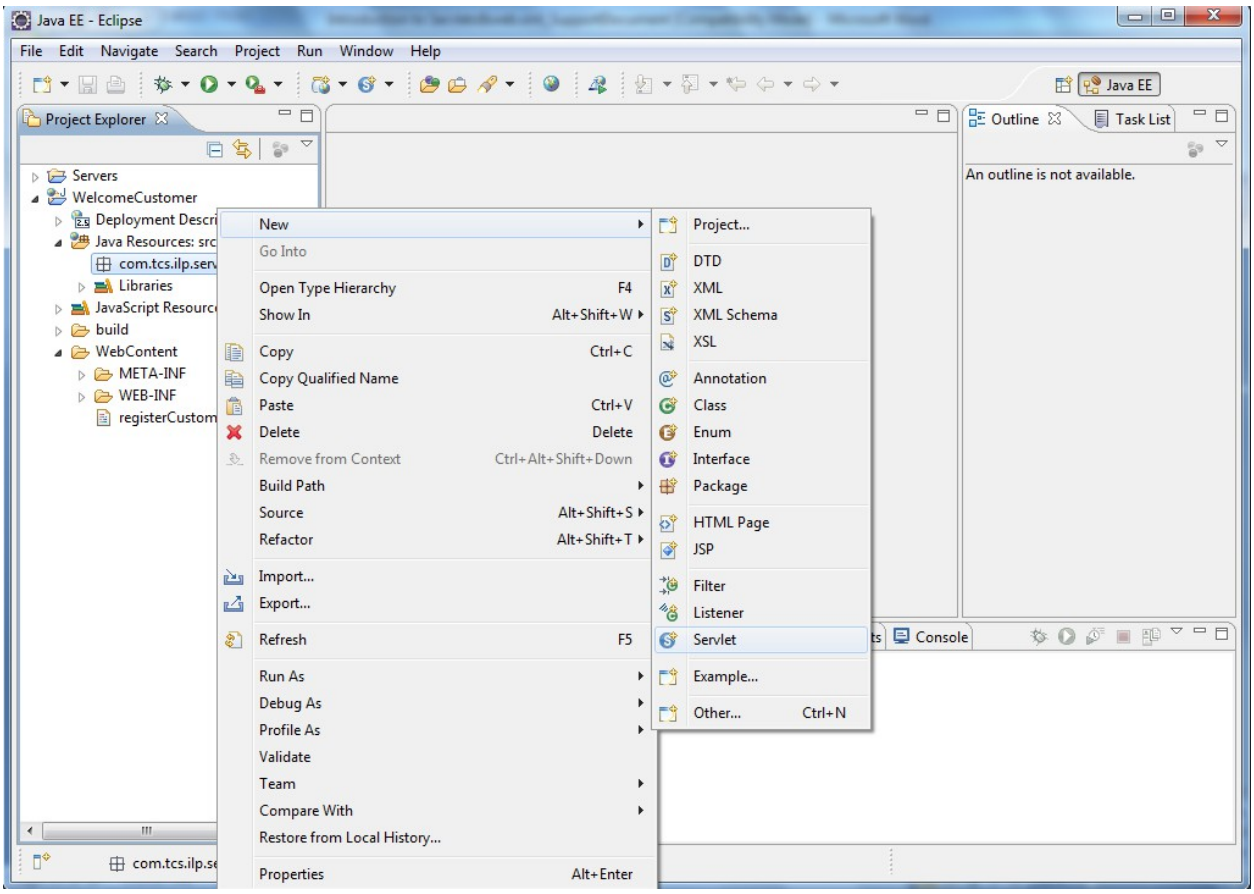
Create a package com.tcs.ilp.servlet.examples inside src folder



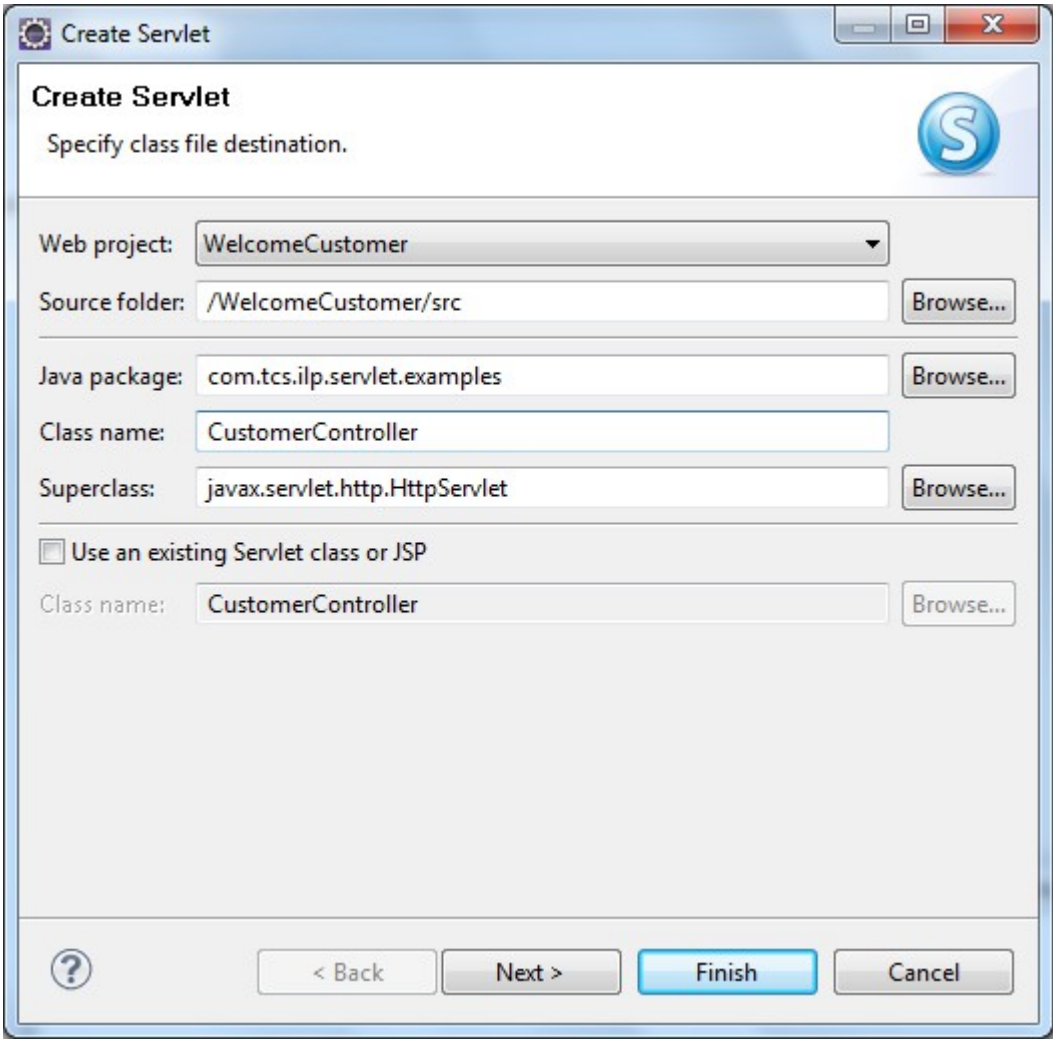
The package gets created inside the src folder



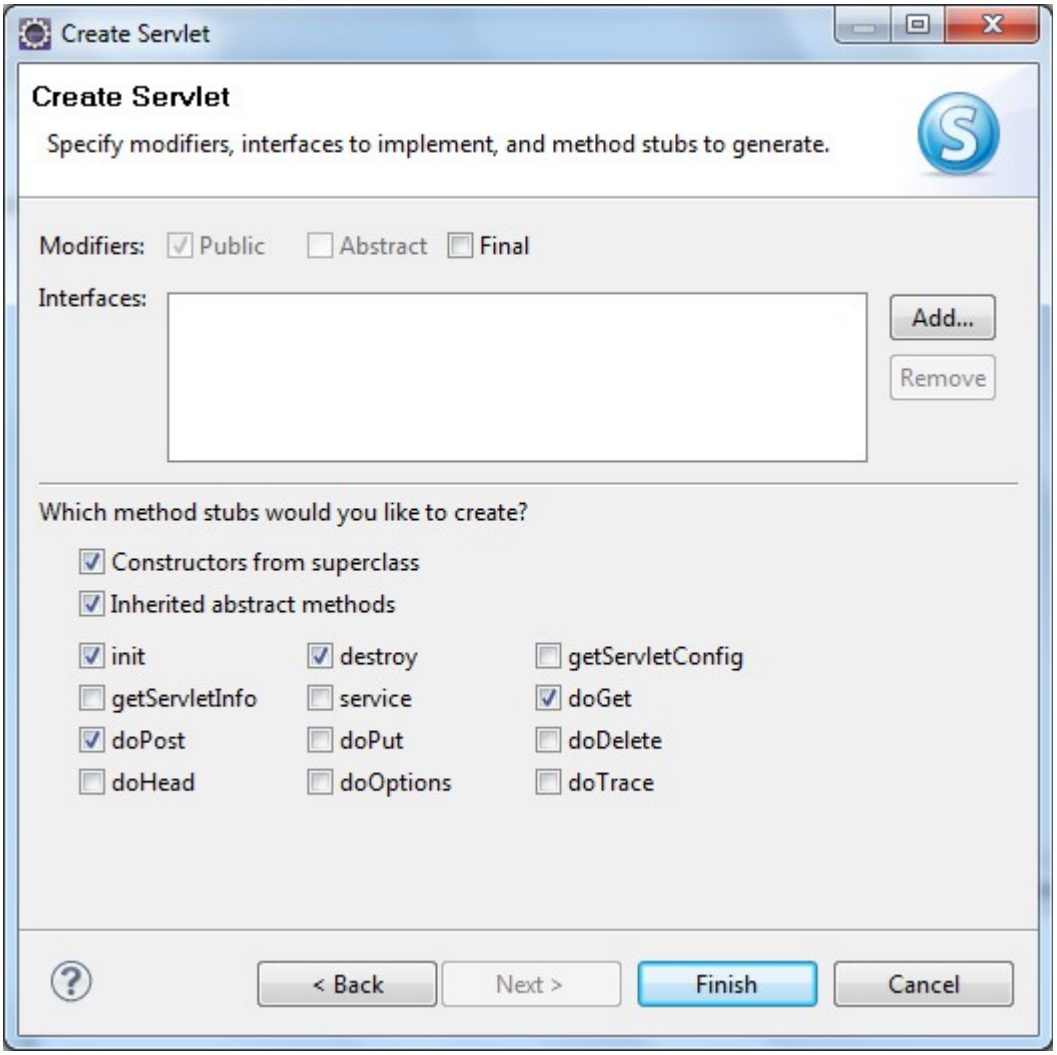
Create a new Servlet CustomerController inside the package.



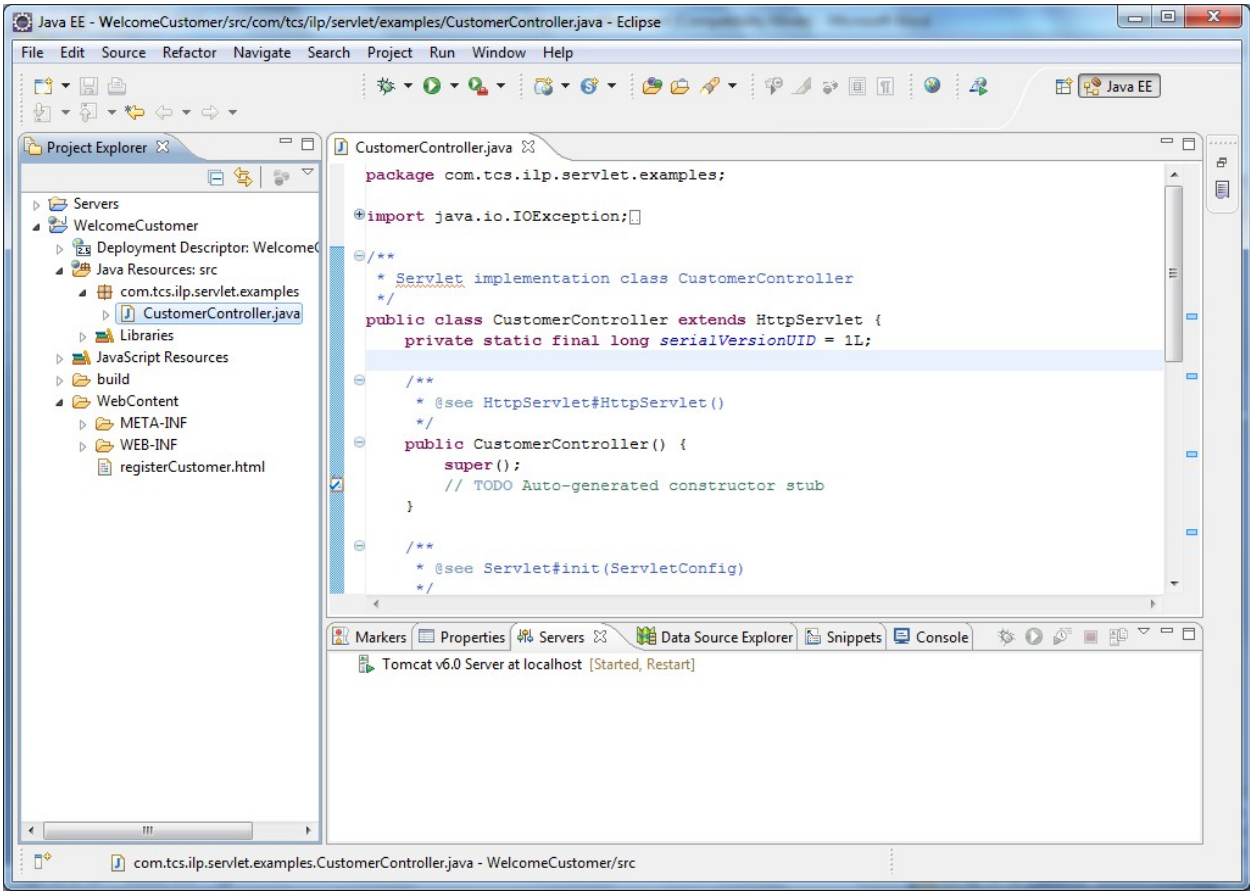
Provide a valid classname for the Servlet and click on Next



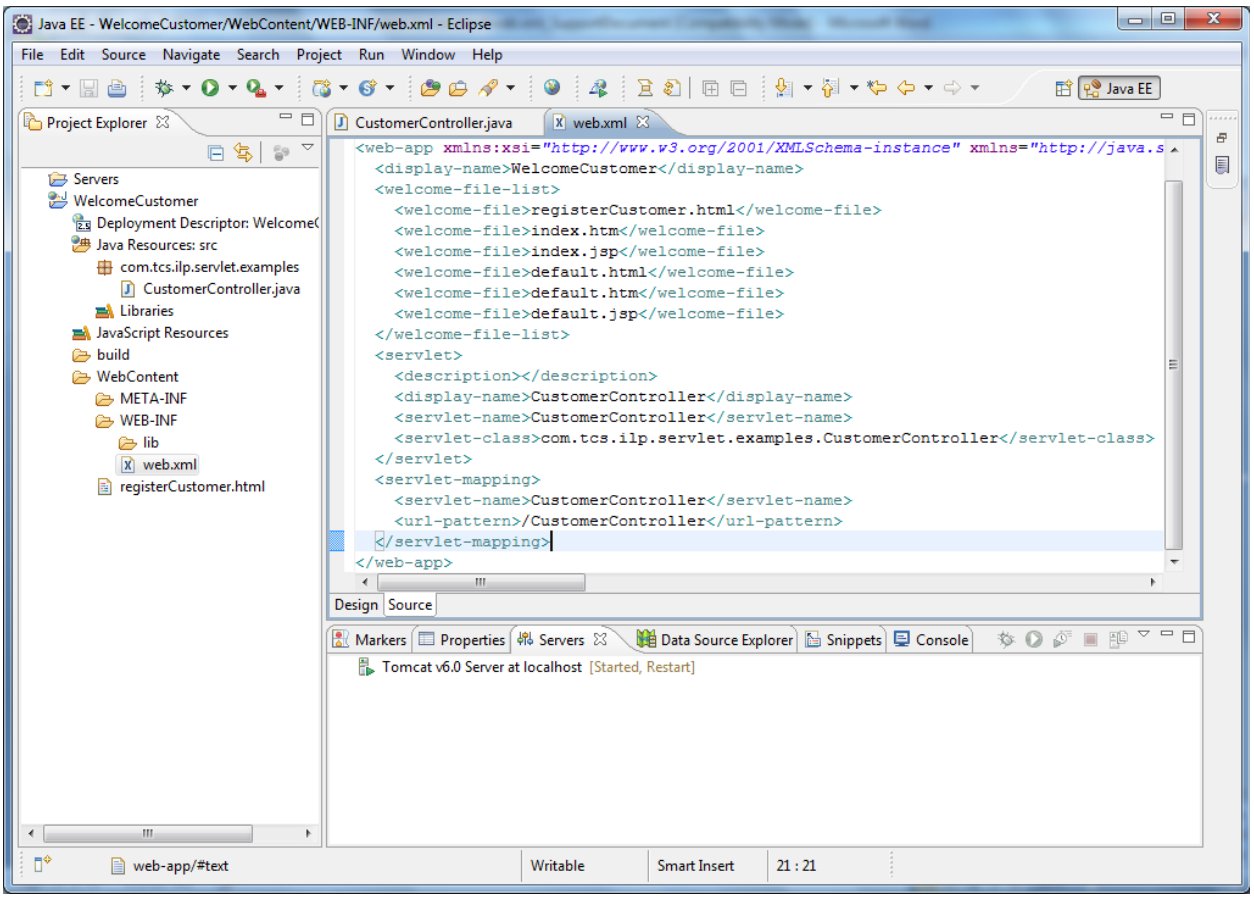
Try to override the lifecycle methods init, destroy,doGet and doPost and click on Finish



The servlet gets created under the package structure



As soon as the servlet is created the corresponding servlet mapping gets created inside the web.xml
Servlet definition has the display name, servlet name and the servlet-class it which it is available
Servlet mapping has the servlet name and the url-pattern.



Code for retrieving the customer keyed in name through request is provided inside servlet

Java EE - WelcomeCustomer/src/com/tcs/ilp/servlet/examples/CustomerController.java - Eclipse

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CustomerController.java

```
package com.tcs.ilp.servlet.examples;

import java.io.IOException;
import java.io.PrintWriter;

import javax.servlet.ServletConfig;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 * Servlet implementation class CustomerController
 */
public class CustomerController extends HttpServlet {
    private static final long serialVersionUID = 1L;

    /**
     * @see HttpServlet#HttpServlet()
     */
    public CustomerController() {
        super();
        // TODO Auto-generated constructor stub
    }

    /**
     * @see Servlet#init(ServletConfig)
     */
    public void init(ServletConfig config) throws ServletException {
        // TODO Auto-generated method stub
        System.out.println("Inside Servlet initilisation");
    }
```

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Java EE - WelcomeCustomer/src/com/tcs/ilp/servlet/examples/CustomerController.java - Eclipse

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CustomerController.java

```
    /**
     * @see Servlet#init(ServletConfig)
     */
    public void init(ServletConfig config) throws ServletException {
        // TODO Auto-generated method stub
        System.out.println("Inside Servlet initilisation");
    }

    /**
     * @see HttpServlet#doGet(HttpServletRequest request, HttpServletResponse
     *      response)
     */
    protected void doGet(HttpServletRequest request,
            HttpServletResponse response) throws ServletException, IOException {
        // TODO Auto-generated method stub
        doPost(request,response);
    }

    /**
     * @see HttpServlet#doPost(HttpServletRequest request, HttpServletResponse
     *      response)
     */
    protected void doPost(HttpServletRequest request,
            HttpServletResponse response) throws ServletException, IOException {
        // TODO Auto-generated method stub
        String customerName= request.getParameter("Name");
        PrintWriter out = response.getWriter();
        out.println("Welcome " + customerName);
    }

}
```

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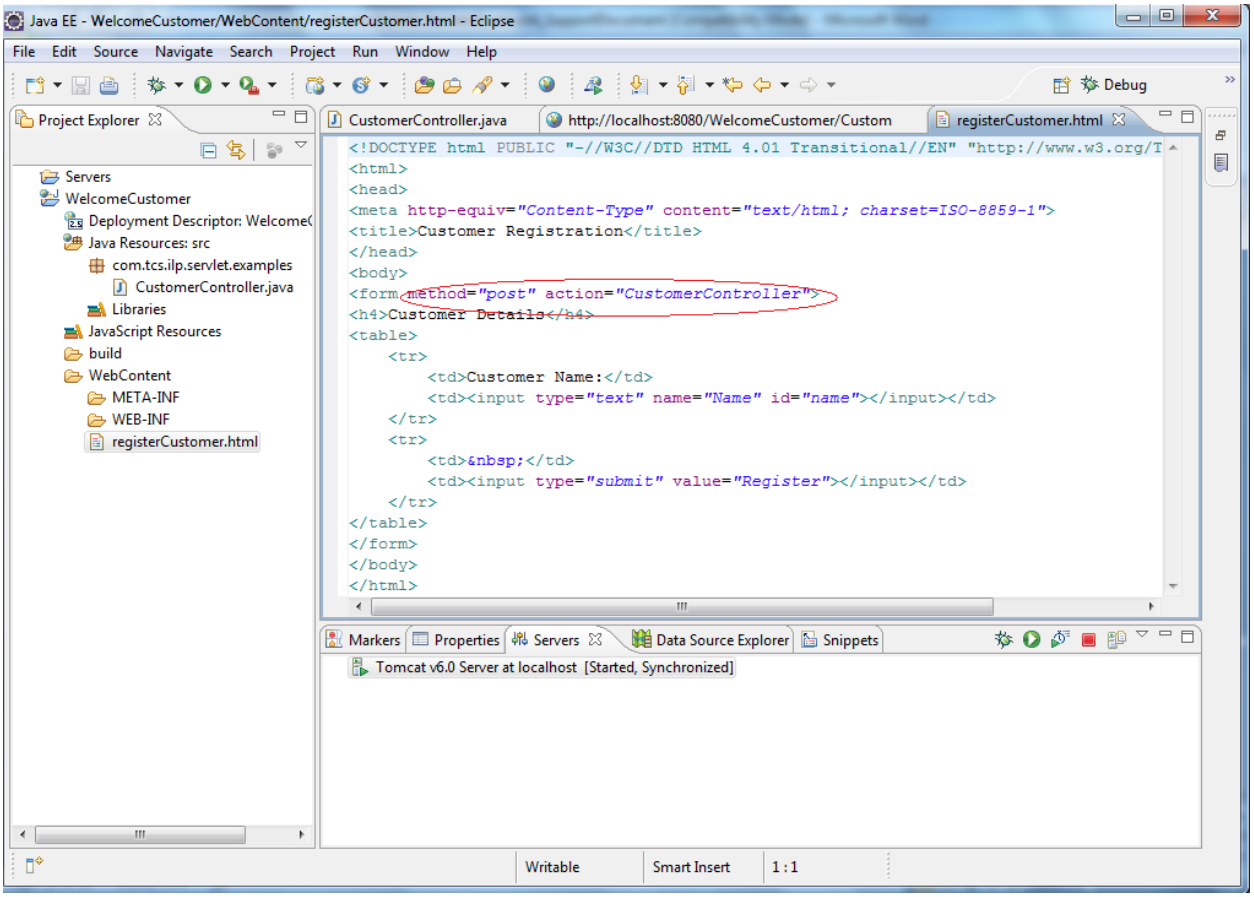
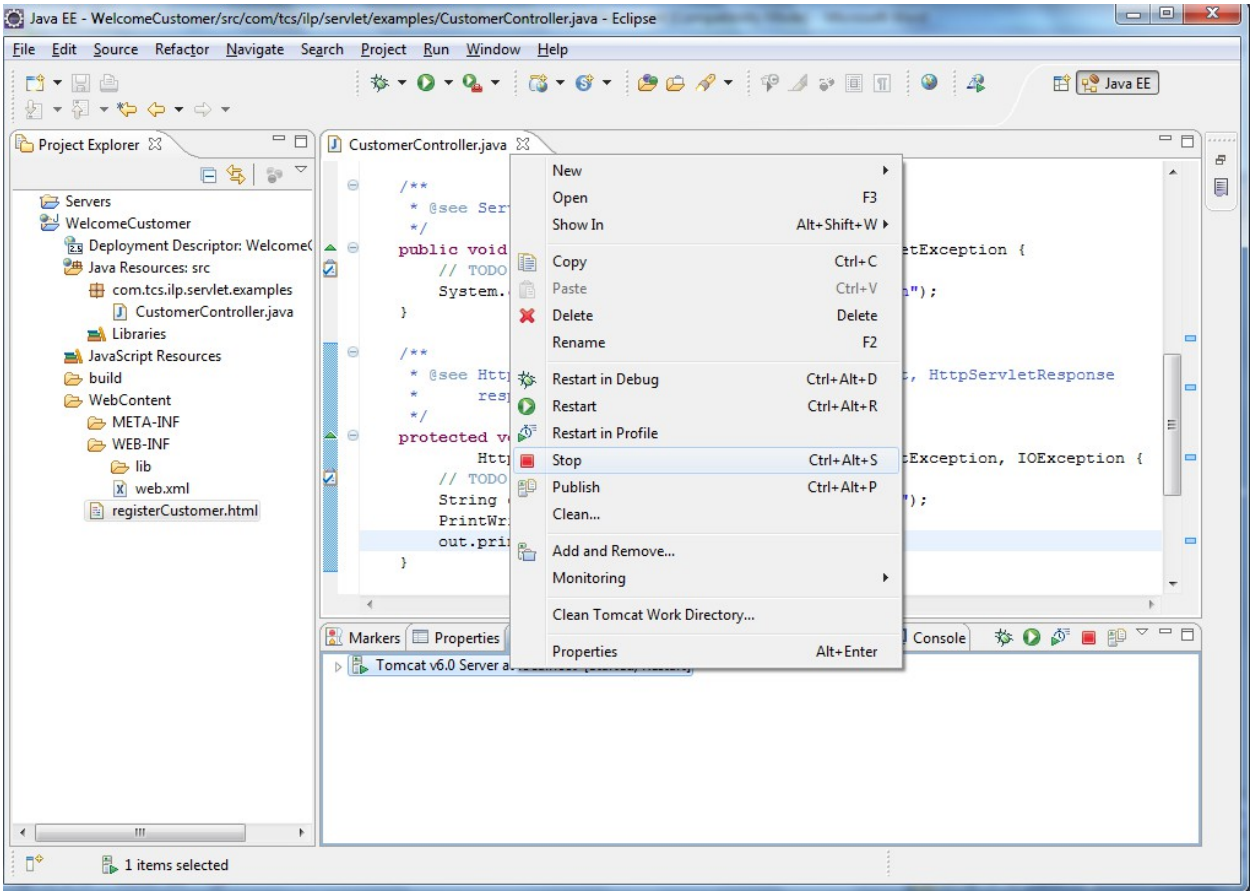
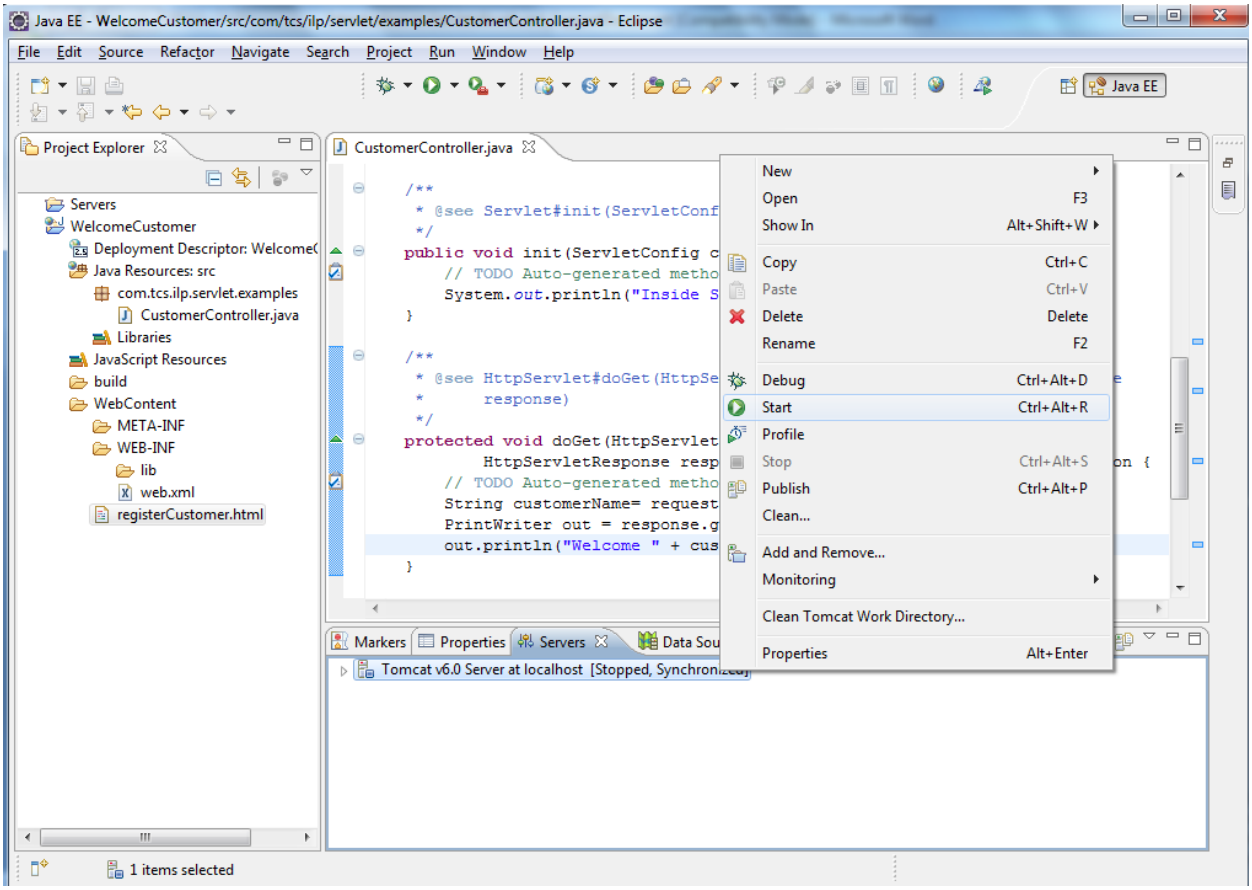
Provide the corresponding action and method inside registerCustomer.html



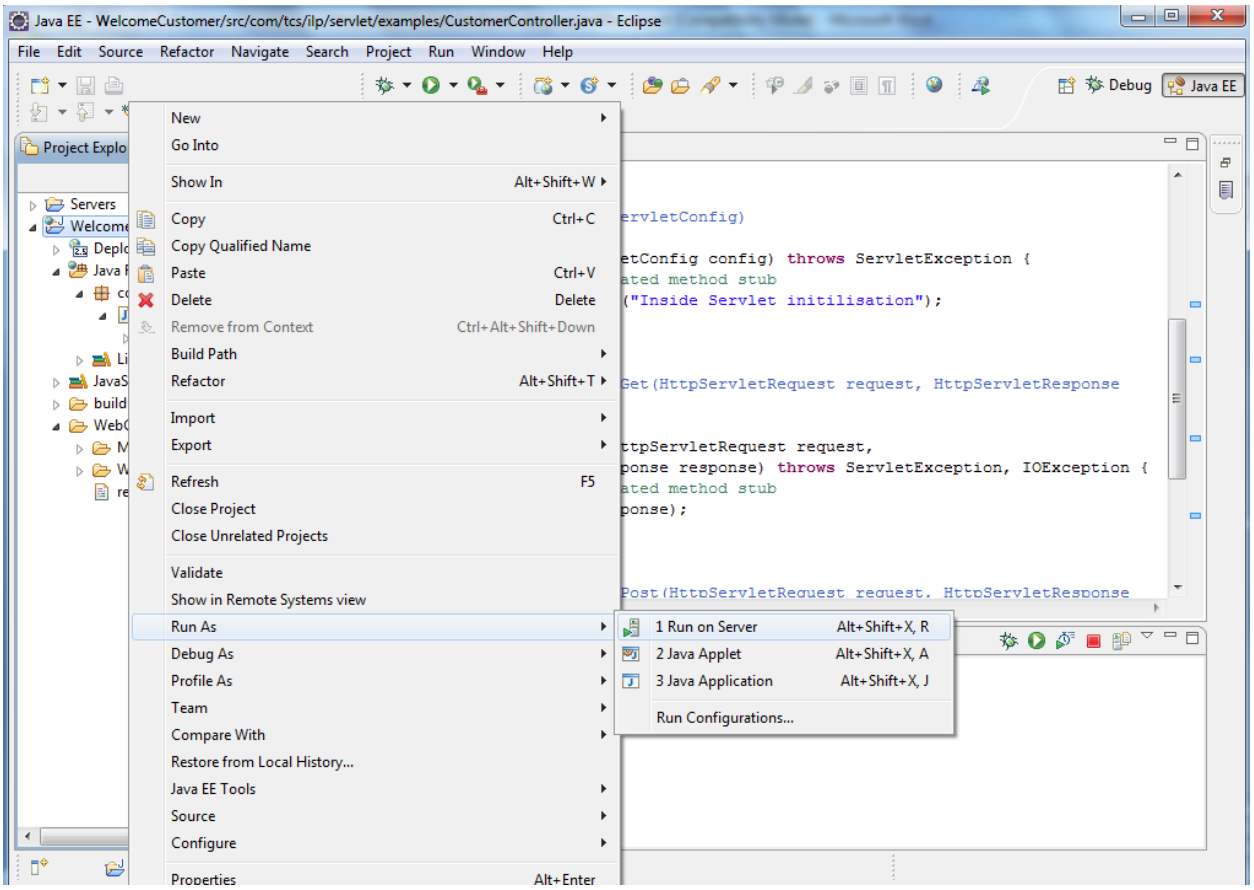
Create the server instance as in WebApplicationBasics_SupportDocument.doc.
Stop the server and restart the server again for the new changes to get reflected



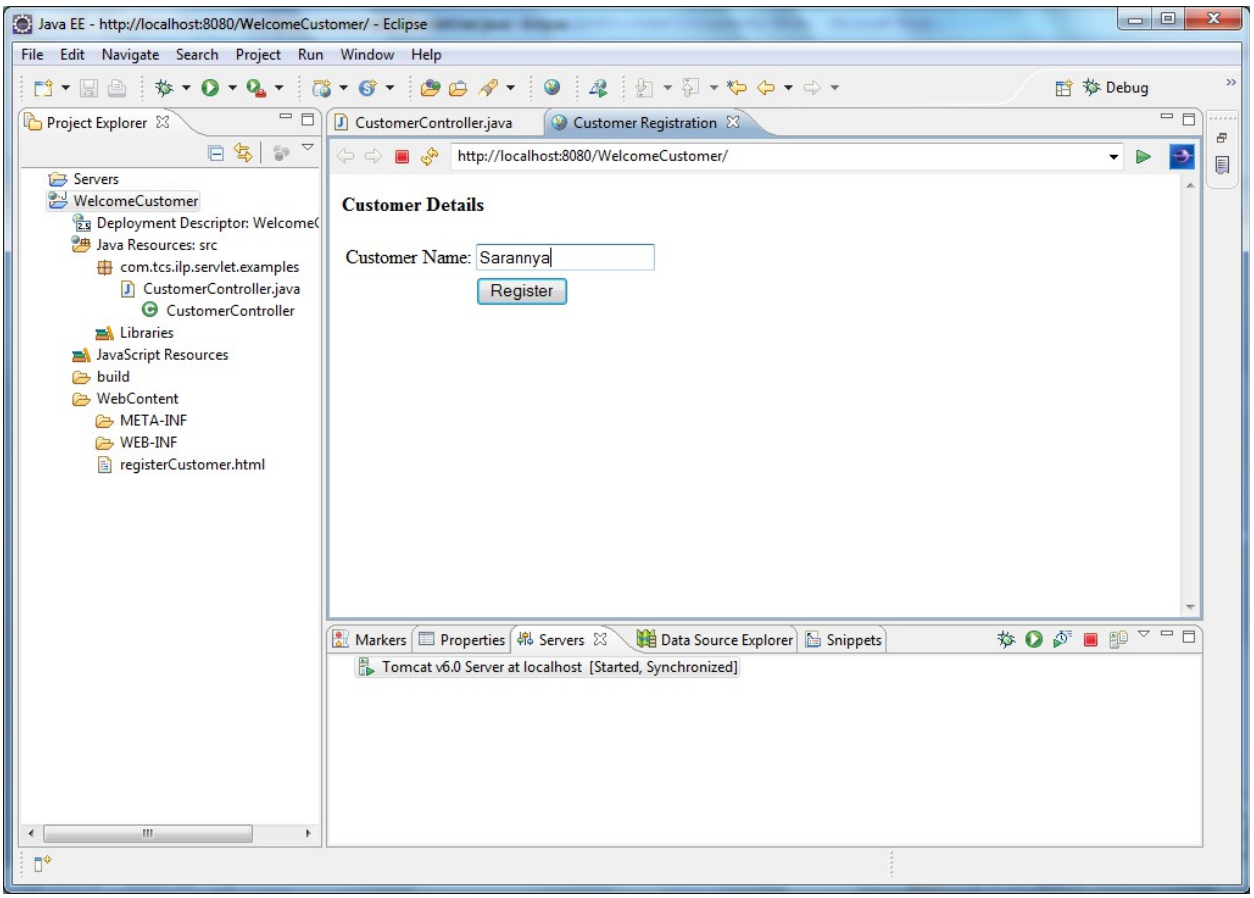
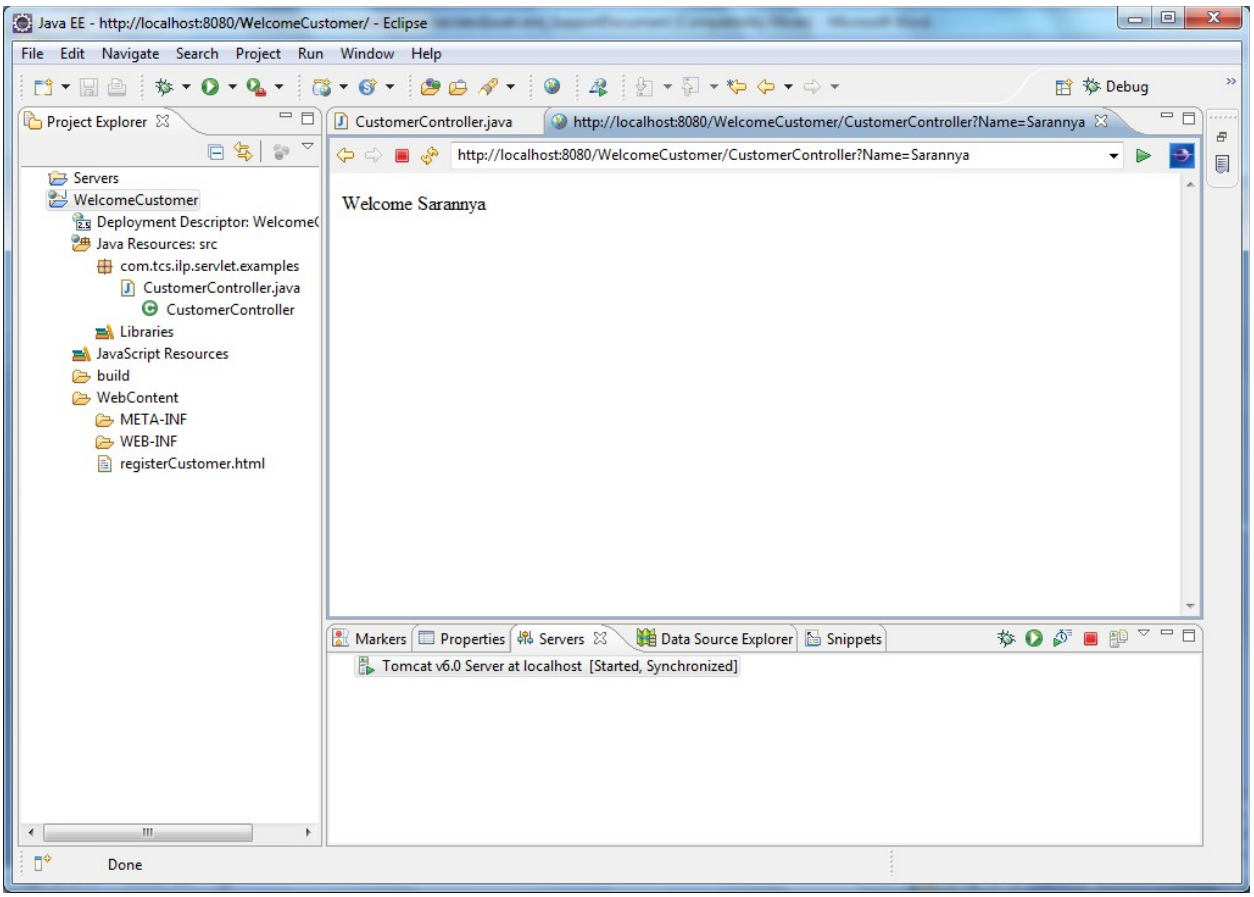
The server will be in stopped status. Now again restart the server.



Once the server gets started, run the dynamic web project on server



The output is shown on the screen. Key in a customer name



A welcome message with customer name gets displayed.